



Fig. 7: HPM e-bomb

cal energy in a time frame of tens to hundreds of microseconds, with corresponding peak power levels of tens of terawatts, is the most mature technology applicable to the design of e-bombs. An e-bomb configured around an FCG is what is known as the low-frequency e-bomb, and its output due to the nature of its basic physics is constrained to a frequency band below 1MHz. FCGs may also be used as one-shot pulse power supplies to drive HPM tubes such as vircators to build an HPM e-bomb.

The basic idea behind the construction of FCG based e-bombs is that of using a fast explosive to rapidly compress a magnetic field, transferring most of the explosive energy into an intense magnetic field. The initial magnetic field in the FCG prior to explosive initiation is produced by a start current, which is supplied by an external source, typically a Marx bank.

Several types of geometrical configurations have been experimented with for building FCGs with coaxial FCG being the most commonly used. Its cylindrical form factor lends itself well to packaging into munitions. The current multiplication ratio of output current to start current is design dependent.

In a munition application, where space and weight are at a premium, the smallest possible start current source is desirable. A current amplification of 60 has been achieved. A cascade arrangement of FCGs may be used in this case.



Fig. 8: Humvee Active Denial System

Here, a small FCG is used to drive a larger FCG with a start current. Fig. 6 shows the component parts of a low-frequency e-bomb that employs a two-stage FCG.

Explosive- and propellant-driven MHD generator technology is another candidate used for generating EMP. An MHD generator depends for its operation on the movement of a conductor through a magnetic field, producing an electrical current transverse to the direction of the magnetic field and conductor motion.

MHD generator technology is not yet mature enough to build practical devices, and there are issues related to the size and weight of magnetic-field-generating devices required for their operation. MHD devices may be used for generating start current for FCG devices. Conductor in an MHD generator is plasma of ionised explosive or propellant gas, which

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